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System Performance Evaluation

Assignment

Report Design

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1 Assignment Description: The AQNet Network and Its Asynchronous Mechanism

1.1 Introduction to the AQNet System

The system our group chose to model and evaluate is AQNet - Asynchronous Queuing Network Simulator. It is a Multi-Hop Packet-Switched Network, works with a Non-blocking Store-and-Forward mechanism.

In modern communication networks, a node (router or switch) may have to wait until the next node is ready before it can send a packet. This can cause deadlock or network-wide congestion. To solve this problem, AQNet uses a non-blocking communication model. Receiving a packet and processing or forwarding it are fully separated. When a packet arrives, the node immediately records it in the output buffer. Then it hands control back to the system (the event loop or scheduler), so the system can keep receiving other packets. The packet waits in the buffer. It is sent only when the output port (the transmission channel) is idle, and this happens asynchronously.

We chose this problem because it is a natural and practical setting. It allows us to combine and verify all six single-queue models required by the project: $M/M/1$, $M/M/1/K$, $M/M/\infty$, $M/M/n/n$, $M/M/c$, and $M/M/c/K$.

1.2 Topology and Packet Flow

AQNet is an open queuing network with no physical feedback loop. It has five functional nodes (N1 to N4 and L1), connected in series and in branches. It also has one independent node, which serves as a reference. A packet travels through the network as follows.

- **Generation at the source (SRC):** Packets are generated independently at `SRC` by a random process. They then enter the network.
- **Hop 1 - Edge router N1 ($M/M/1/K$ model):** The packet enters the edge access node N1. This node has a single transmission port ($c = 1$) and a finite buffer of size K. If a packet arrives when the buffer already holds K packets, it is dropped immediately (tail-drop).
- **Hop 2 — Optical propagation delay L1 ($M/M/\infty$ model):** After N1 sends the packet out, it travels along the physical link L1. On this link, packets travel



independently and do not compete for resources. This is equivalent to having infinitely many servers. The packet only experiences a fixed propagation delay before it reaches the next destination.

- **Hop 3 — Core switch N2 ($M/M/c/K$ model, the bottleneck):** After leaving L1, the packet enters the network core at N2. Here, c parallel service channels share one finite buffer of size K. Heavy traffic converges at this node, so it is considered the main bottleneck of the whole network. It determines the throughput and the highest packet loss rate.

- **Hop 4 — Branching (routing):** After leaving N2, the packet does not follow a single path. The router splits the load by independent probabilities:

  - **Branch A (probability $p = 0.6$)** goes to N3 ($M/M/n/n$ model). N3 acts as a trunking system (or load balancer) with n channels that serve packets directly. It has no waiting buffer. If a packet arrives when all n channels are busy, the packet is rejected immediately (an Erlang loss system).
  - **Branch B (probability $p = 0.4$)** goes to N4 ($M/M/c$ model). N4 is a local access switch with c service channels and a very large buffer, which is assumed to be infinite. Buffer overflow does not occur.

- **End at the destination (SNK):** Packets that get through N3 or N4 reach the destination `SNK`. They complete their life cycle and leave the system.

- **Baseline model ($M/M/1$):** To isolate the system for theoretical analysis, we design a separate $M/M/1$ node with an infinite buffer. It runs in parallel with the network. It serves as a reference for comparing delay and throughput in the ideal case, when there is no congestion.

1.3 Asynchronous Features and Task Lifecycle Management

To keep the system non-blocking, all processing in AQNet is carried out through a discrete-event simulation (DES) Scheduler rather than by waiting on the system (wall-clock) time. Each packet is managed by an asynchronous task throughout its lifecycle. The table below maps the asynchronous points of the real problem to the modeling mechanisms in AQNet.

Asynchronous Requirement (Real-world Operation)	**Processing & Mapping Mechanism in the AQNet Model**
Decoupling Reception from Processing (Non-blocking I/O)	The Dispatcher inserts each "packet arrival" event into the Priority Queue and immediately returns control, so network nodes are never blocked while waiting for a transmission port. Virtual Time advances only when the Scheduler dequeues the next event for processing, independent of wall-clock time.
Message Buffering and Order Preservation	Nodes `N1`, `N2`, and `N4` each maintain a dedicated State Queue to hold packets that have not yet been served. The queues follow a strict FCFS (First-Come, First-Served) discipline, ensuring packets are served in order of arrival and thereby preserving causality in the simulation.
Possibility of Message Loss (Packet Loss)	A packet may be lost in two cases: 1. **Buffer overflow** at nodes with finite queues (`N1`, `N2`): an arriving packet is discarded when the queue is full (Blocking / Tail-drop). 2. **Immediate rejection** at the bufferless node (`N3`): if the node is busy upon arrival, the packet is dropped instantly (Erlang Loss).
Waiting Time Limit (Timeout/TTL)	Each packet is assigned a maximum lifetime (Time-To-Live, TTL). Measured from its entry into the network at `N1`, if the elapsed time exceeds the TTL before the packet reaches the `SNK`, the system **cancels the packet** (Cancelled/Timeout) to release network resources, emulating the Drop Timeout mechanism of IP packets.

Table 1.1: Mapping of asynchronous requirements to modeling mechanisms in AQNet

2 Defining the System Boundary, Services, and Outcomes

2.1 System Boundary Diagram

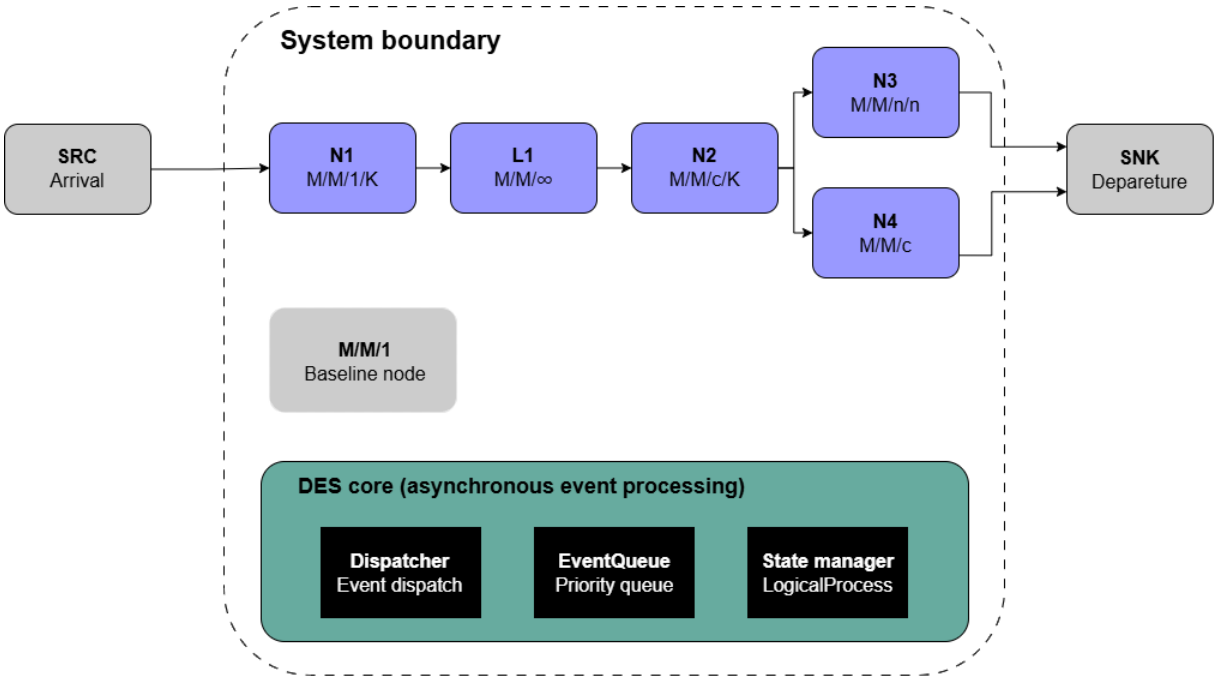


Figure 2.1: System Boundary Diagram

2.2 Service - Outcomes - Corresponding Metrics Table

Service	**Code**	**Possible Outcome**	**Cause**	**Corresponding Metric**
Asynchronous receiving, buffering, and forwarding of packets through the AQNet network (Store-and-Forward)	O1	**Completed**	The packet passes successfully through all network nodes (`N1`, `L1`, `N2`, and `N3`/`N4`) and safely reaches the Sink.	M01 (ρ), M02 (L), M03 (L_q), M04 (W), M05 (W_q), M06 (X), M08 (distribution p_n), M09 (E2E delay)
	O2	**Blocked** (buffer full, finite-K systems)	The packet arrives at a node with a finite queue (`N1` – $M/M/1/K$, or `N2` – $M/M/c/K$) when the buffer already holds the maximum of K packets, so it is dropped.	M07 (P_{block})
	O3	**Rejected** ($M/M/n/n$, all channels busy)	The packet is routed to the trunking node `N3`, but all n channels are busy. There is no waiting space, so the service is refused (Erlang loss).	M07 (P_{block})
	O4	**TTL expired / Timeout**	The packet's time in the network exceeds the allowed TTL before it reaches the destination, so the scheduler cancels it.	M10 (TTL drop rate)
	O5	**Communication error / Cancelled**	The event-processing task is deliberately cancelled, or an error occurs in the Future/TaskGroup communication mechanism.	M11 (counter that records error cases, as defined in Design 2)

Table 2.1: Mapping of services, outcomes, and metrics in AQNet

3 Mapping Real-World Parameters to Queueing Parameters

3.1 Mapping Table of Real-World Parameters to Queueing Parameters

To set the input parameters for the simulation, we convert the real network into quantities from queueing theory.

Converting real-world figures to get $\mu = 100$ pkt/s (standard H6):

- Assume the average packet size in the network is $\bar{S}$ = 1,000 bytes = 8,000 bits (exponentially distributed).
- The bandwidth of one standard port or transmission channel is B = 800 kbps = 800,000 bits/s.
- Then the average service rate of one channel is $\mu = \dfrac{B}{\bar{S}} = \dfrac{800{,}000}{8{,}000} = 100$ packets/s (pkt/s). The average service time of one packet is $1/\mu = 10$ ms = 0.01 s.

Queueing Parameter	**Real-world Quantity in AQNet**	**Unit**	**Value / Range**	**Modeling Assumption**
λ (arrival rate)	The rate at which packets are generated by the source (`SRC`) and enter the network.	pkt/s	Varies with the load level $\rho \in \{0.3, 0.6, 0.9\}$.	The interarrival time follows an exponential distribution (Poisson arrival process).
μ (service rate)	The capacity of one network port or one hardware channel to transmit and process packets.	pkt/s	Normalized to $\mu = 100$ pkt/s.	The packet size (equivalent to the service time) follows an exponential distribution.
c or n (number of servers)	The number of parallel output ports at the switches `N2` and `N4` (c), or the number of trunk channels at `N3` (n).	ports / channels	$c = 4$ (at `N2`, `N4`); $n = 5$ (at `N3`).	The parallel ports in the $M/M/c$ model share one common queue (Shared Queue) instead of separate queues.
K (system capacity)	The buffer capacity at a network port, measured as the maximum number of packets it can hold, including the packet being transmitted.	packets	$K \in \{10, 20\}$ at the finite nodes.	A packet that arrives when the buffer is full is dropped immediately (Tail-drop). `N3` has no buffer ($K = n$).
SD (service discipline)	The scheduling rule for processing and forwarding packets in the buffer.	—	FCFS (First-Come, First-Served).	It keeps packets in their order of arrival, as the non-blocking communication system requires.
Topology & p_{ij}	The network diagram and the load-splitting (routing) probabilities at the routers.	—	$p_{23} = 0.6$ (via branch A); $p_{24} = 0.4$ (via branch B).	Routing is probabilistic and independent. It does not depend on the instantaneous congestion of the next nodes.

Table 3.1: Mapping of AQNet physical quantities to queueing parameters

3.2 Discussion of the Markov Assumption in the Model

All six required single-queue models in this project are Markovian ($M/M/\dots$). This means arrivals follow a Poisson process and service times follow an exponential distribution. This assumption gives the system the *memoryless* property.

- **Justification and benefits:** The Markov assumption is necessary in Phases 1 and 2. It allows us to validate the simulation results against closed-form analytical formulas, such as Erlang B, Erlang C, and the product-form solution of Jackson networks. Without the memoryless property, we could not use Markov chains to compute the reference data.
- **Implications for real-world use:** Real Internet traffic is often bursty and self-similar. Packet sizes often follow heavy-tailed (Pareto) distributions rather than the exponential distribution. As a result, during traffic bursts, the analytical model tends to underestimate the real queue length and the packet loss probability (P_{block}).
- **Design approach:** Although we use the Markov model as a baseline, the AQNet DES architecture remains flexible. The `ArrivalSource` and `Server` classes accept any specified distribution, so we can run experiments with self-similar traffic when needed.

4 Queuing Model and Assumptions

AQNet (Asynchronous Queuing Network Simulator) represents a multi-hop packet-switched network with non-blocking store-and-forward operations. A node receives a packet, stores it if necessary, and schedules later processing rather than synchronously waiting for forwarding to finish. The queuing model provides a mathematical abstraction of that behaviour.

4.1 Queuing Models Used in AQNet

Table 4.1 summarises six queuing models and their interpretation in AQNet. The M/M/1 model provides an isolated baseline rather than an additional network node.

Table 4.1: Queuing models and their roles in AQNet

Model	Interpretation in AQNet
M/M/1	One output port with an effectively unlimited waiting queue (baseline).
M/M/1/K	Edge router N1 with one server and finite system capacity.
M/M/∞	Propagation stage L1 with no contention for servers.
M/M/n/n	N3: n service channels, no waiting room; reject if all are busy.
M/M/c	N4: c service channels with a shared waiting queue.
M/M/c/K	N2: c service channels with finite system capacity.

4.2 Notation and Model Assumptions

Let λ_i denote the mean packet arrival-attempt rate at node i (packets/s), μ_i the mean service rate of one server (packets/s), c_i the number of identical servers, and K_i the maximum number of packets in the node, including those in service. The mean service time is $1/\mu_i$. A finite-server offered-load ratio is

$$\rho_i = \frac{\lambda_i}{c_i \mu_i}, \qquad a_i = \frac{\lambda_i}{\mu_i}. \tag{4.1}$$

For one server, $c_i = 1$. For M/M/∞, a_i is an offered load, not a finite-server utilisation ratio.

The isolated M/M formulas assume Poisson arrival attempts and exponentially distributed service times. FCFS and a common waiting queue are assumed for multi-server

waiting models. These are *analytical assumptions*, not claims about every traffic stream inside the coupled AQNet network. In particular, a finite buffer or TTL expiration can alter traffic seen by downstream nodes. Throughout the analysis, K_i denotes total system capacity, including service positions; this convention must also be used when interpreting simulation results.

5 AQNet Topology and Network Equations

5.1 Network Topology

The network configuration considered in this analysis consists of SRC → N1 → L1 → N2. A packet forwarded from N2 goes to N3 with probability 0.6 or to N4 with probability 0.4, and then reaches SNK if successfully delivered (Figure 5.1). An isolated M/M/1 system is used separately as a baseline.

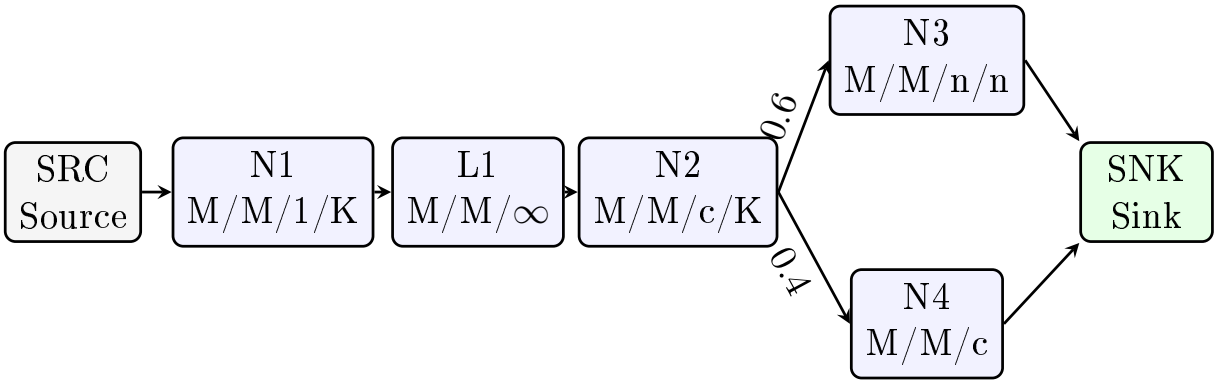


Figure 5.1: AQNet network model and routing probabilities.

Figure 5.1 represents the feed-forward configuration used in this analysis. Node-specific parameters are defined separately, and feedback links are not included in this configuration.

5.2 Traffic Flow Equations

For an ideal lossless open network, the standard flow equation is

$$\lambda_i = \gamma_i + \sum_j \lambda_j p_{ji}, \tag{5.1}$$

where γ_i is the external arrival rate and p_{ji} is the routing probability from node j to node i.

For a network with rejection, distinguish the *offered rate* λ_i from the *successful forwarding rate* X_i:

$$\lambda_i = \gamma_i + \sum_j X_j p_{ji}. \tag{5.2}$$

For the feed-forward AQNet configuration this becomes

$$\lambda_{N1} = \lambda_{\text{SRC}}, \qquad \lambda_{L1} = X_{N1}, \tag{5.3}$$

$$\lambda_{N2} = X_{L1}, \qquad \lambda_{N3} = 0.6X_{N2}, \tag{5.4}$$

$$\lambda_{N4} = 0.4X_{N2}, \qquad X_{\text{SNK}} = X_{N3} + X_{N4}. \tag{5.5}$$

These equalities assume that successful downstream forwarding follows the illustrated routes and that packets are not duplicated. Additional routes or feedback links require additional flow terms.

5.3 Admission, Blocking and Stability

Let $P_{\text{block},i}$ be the probability that a packet offered to node i is rejected for lack of capacity. The admitted rate is

$$\lambda_{\text{eff},i} = \lambda_i(1 - P_{\text{block},i}). \tag{5.6}$$

If all admitted packets complete service without further failures, $X_i = \lambda_{\text{eff},i}$. If packets expire or fail after admission, then X_i may be smaller. Under the isolated Poisson-arrival assumptions, M/M/1/K and M/M/c/K have $P_{\text{block}} = p_K$, while M/M/n/n uses Erlang B:

$$P_{\text{block}} = \frac{a^n/n!}{\sum_{k=0}^{n} a^k/k!}, \qquad a = \lambda/\mu. \tag{5.7}$$

These are not automatically exact for downstream AQNet traffic altered by earlier packet losses.

For infinite-waiting-room M/M/1 and M/M/c models, the familiar steady-state conditions are, respectively,

$$\lambda < \mu, \qquad \lambda < c\mu \quad \text{(M/M/c)}. \tag{5.8}$$

Thus the M/M/c node N4 requires $\lambda_{N4} < c_{N4}\mu_{N4}$ under its model assumptions. Finite systems N1, N2 and N3 have bounded occupancy and can have a stationary distribution even at offered load ratios of one or above, although blocking may become substantial.

The ideal M/M/∞ stage has no server-contention waiting.

5.4 End-to-End Delay

For a packet k successfully delivered from SRC to SNK, define

$$D_k = t_{k,\mathrm{SNK}} - t_{k,\mathrm{SRC}}. \tag{5.9}$$

The reported mean is calculated over successful deliveries:

$$D_{\mathrm{E2E}} = \frac{1}{N_{\mathrm{success}}} \sum_{k \in \mathcal{S}} D_k, \quad N_{\mathrm{success}} = |\mathcal{S}| > 0. \tag{5.10}$$

For a simplified *lossless* path calculation, let W_i be the mean sojourn time at node i. The two possible paths have

$$D_3 = W_{N1} + W_{L1} + W_{N2} + W_{N3}, \tag{5.11}$$

$$D_4 = W_{N1} + W_{L1} + W_{N2} + W_{N4}. \tag{5.12}$$

If successful deliveries retain the routing proportions, their mean is

$$D_{\mathrm{E2E}} = 0.6D_3 + 0.4D_4. \tag{5.13}$$

With unequal branch success probabilities g_3, g_4, the routing weights must instead be conditioned on successful delivery:

$$q_3 = \frac{0.6g_3}{0.6g_3 + 0.4g_4}, \quad q_4 = 1 - q_3, \tag{5.14}$$

provided the denominator is nonzero. The corresponding mean is $q_3 D_{3|\mathrm{success}} + q_4 D_{4|\mathrm{success}}$.

5.5 Analytical Validity and Jackson Network Limitations

The classical open Jackson network provides a product-form steady-state solution under its specified assumptions. The AQNet model additionally includes finite-capacity rejection and potential TTL/timeout termination. These can change downstream arrival streams and create dependencies between nodes. Therefore the classical Jackson product-form solution must *not* be assumed exact for AQNet as a whole. Isolated-node M/M formulas remain analytical references only when their individual assumptions apply.

A full buffer may cause packet dropping or upstream backpressure, depending on the forwarding mechanism. The blocking equations above represent admission rejection, whereas a backpressure model requires additional treatment of upstream dependencies. The M/M/∞ stage idealises propagation using exponential service times and no server contention, rather than an exact physical link-delay model.

5.6 Model Consistency Conditions

The following conditions support consistency checks between the mathematical model and simulation results:

$$p_{N2,N3} + p_{N2,N4} = 0.6 + 0.4 = 1, \tag{5.15}$$

$$0 \leq N_i(t) \leq K_i \quad \text{(finite-capacity node)}, \tag{5.16}$$

$$L_i = \lambda_{\text{eff},i} W_i \quad \text{(consistent steady-state population)}. \tag{5.17}$$

Packet outcomes should also be counted without overlap:

$$\begin{aligned} N_{\text{generated}} = N_{\text{delivered}} + N_{\text{rejected}} \\ + N_{\text{terminated}} + N_{\text{remaining}}. \end{aligned} \tag{5.18}$$

The conservation check assumes every generated packet appears in exactly one final status category at the observation end. Little's Law also requires consistent arrival/exit populations, especially if packets terminate after admission.

5.7 Scope of the Analytical Model

The equations describe the feed-forward routing configuration in Figure 5.1, with K_i defined as total capacity. Node-specific arrival and service parameters determine numerical performance estimates. TTL expiration is treated as a separate packet-termination outcome; its threshold and triggering rule are model parameters. The formulas are analytical relationships, not measured network results.